DIMENSION LOWER BOUND FOR COMPONENTS OF AN INTERSECTION IN A NONSINGULAR VARIETY

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0B0Q: sources/stacks-chow/tags.txt, source lines 13868–13868. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0AZN: sources/stacks-chow/tags.txt, source lines 13831–13831. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a nonsingular variety and let $W, V \subset X$ be closed subvarieties with $\dim(W) = s$ and $\dim(V) = r$. Then every irreducible component Z of $V \cap W$ has dimension $\geq r + s - \dim(X)$.*

Proof. Since $V \cap W = \Delta^{-1}(V \times W)$ (scheme theoretically) we conclude by Lemmas 0B0Q and 0AZN. □

Date: stacks-project:intersection.tex:local-snapshot-20261012; GNU FDL 1.2 or later.